Introduction of Financial Professional Certificate



Fixed Income

Goals



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Introduction to Fixed Income Securities

What is a fixed Income Security?

An investment that provides a **return** in the form of **fixed periodic payments** and the eventual return of **principal at maturity**.

Unlike a variable-income security, where payments change based on some underlying measure such as short-term interest rates, the payments of a fixed-income security are known in advance.

Example

a **$ 1000** US Treasury bond which pays a **6% annual coupon** with **five years** maturity.

Types of Fixed Income

FIS	Issuer	Maturity(years)	Remarks
Treasury Bills	US Treasury	< 1	Effectively zero coupon bonds
Treasury Notes	US Treasury	2,3,5,10	Prices quoted in percent and 32nd of 1% face value
Treasury Bonds	US Treasury	20/30	Non callable
Treasury Inflation Protected Securities	US Treasury	5,10,20	Par value continually adjusted based on inflation level
Treasury Strips	Various		Formed by converting Tresury securities into zero coupon bonds
Agency Bonds	Federally related Institutions(Ginnie Mae), Government Sponsored Enterprises(Freddie Mac)		
Mortgage- backed Securities	Ginnie Mae, Fannie Mae, Freddie Mac		
Municipal Bonds	State and Local Government		Most bonds coupon interest payment is tax exempt
Tax Backed Bonds	State and Local Government		
Revenue Bonds	State and Local Government		Payments made only through the revenue generated

Common Terms Explained

Par Value	It is the face value of a bond. It is also the price of a bond when the coupon rate equals to the Yield to measure rate
Coupon	The interest rate stated on a bond when it is issued. The coupon is typically paid semiannually
Maturity	Upon maturity of a fixed income investment such as a bond, the borrower has to pay back the full amount of the outstanding principal, plus any applicable interest to the lender
Zero coupon bonds	A debt security that doesn't pay interest (a coupon), rendering profit at maturity when the bond is redeemed for its full face value

Time Value of Money

- The idea that money available at the present time is worth more than the same amount in the future due to its potential earning capacity.

- This core principle of finance holds that, provided money can earn interest, any amount of money is worth more the sooner it is received.

Valuation of Bonds : Annual Coupon Bonds

- Consider a security that will pay $ C per year for ten years and make a single $ P payment at maturity. The value of bond is calculated by discounting the cash inflows with a discounting rate (say d%):

$$\text{Value of Bond} = \frac{C}{1+d} + \frac{C}{(1+d)^2} + \frac{C}{(1+d)^3} + \cdots . + \frac{C}{(1+d)^9} + \frac{C}{(1+d)^{10}} + \frac{P}{(1+d)^{10}}$$

Example 1

- A $1000, 7% 10 year annual pay bond has a yield of 7.8%. If the yield remains unchanged, how much will the bond value increase over the next 4 years?

a) $16.96

b) $17.25

c) $ 17.89

d) $ 16.34

Valuation of Bonds : Semi Annual Coupon Bonds

- Consider a security that will pay $ C semi annually per year for ten years and make a single $ P payment at maturity. The value of bond is calculated by discounting the cash inflows with a discounting rate (say d%):

$$\text{Value of Bond} = \frac{C/2}{1+d/2} + \frac{C/2}{(1+d/2)^2} + \frac{C/2}{(1+d/2)^3} + \cdots . + \frac{C/2}{(1+d/2)^{19}} + \frac{C/2}{(1+d/2)^{20}} + \frac{P}{(1+d/2)^{20}}$$

Valuation of Bonds : Zero Coupon Bonds

- No coupon bonds, only payment of principal at maturity

- Consider a zero coupon bond that will pay a single $ P payment at maturity. The value of bond is calculated by discounting the cash inflow with a discounting rate (say d%):

$$\text{Value of Bond} = \frac{P}{(1+d)^{10}}$$

Example 2

- An investor buys a 10 year $10,000, 8% coupon, semiannual pay bond for $9,100. He sells it four years later, just after receiving the eighth coupon payment, when its yield to maturity is 5.6%. What would be the bond price at the time of sale?

a) $ 10,563

b) $ 11,209

c) $ 12,234

d) $ 13,983

Traditional Yield Measures

- Current Yield: the annual interest income from the bond

Current Yield = Annual Coupon interest received/Bond Price

- The current yield is simply the coupon payment (C) as a percentage of the (current) bond price (P).

$$\text{Current yield} = C / P_0.$$

- Drawbacks :

- Only Considers coupon interest
- Capital Gains/Losses not taken into account

Traditional Yield Measures

- Yield to Maturity(YTM): YTM is the IRR of the bond. It is the annualized rate of return on the bond

$$\frac{C}{\left(1+\frac{YTM}{2}\right)}+\frac{C}{\left(1+\frac{YTM}{2}\right)^{2}}+.....+\frac{(C+Par)}{\left(1+\frac{YTM}{2}\right)^{2N}}$$

- Yield Measure Relationships:

Bond Selling at:	Relationship
Par	Coupon rate = Current Yield = Yield to Maturity
Discount	Coupon rate < Current Yield < Yield to Maturity
Premium	Coupon rate > Current Yield > Yield to Maturity

- Advantages:

•Considers both coupon income and capital gain/loss if held to maturity.

•Considers the timing of cash flows

Traditional Yield Measures

- YTM of Annual Coupon Bond:
- A 10 year, $1000 par value bond has a coupon of 7%. If it is priced at $920 what is the YTM?
- PV = -920; N=10; FV=1000; PMT=70
- I/Y = 8.20%

Traditional Yield Measures

- Bond Equivalent Yield (BEY): allows fixed-income securities whose payments are not annual to be compared with securities with annual yields.

  –yields are stated at a semi annual rate, it is then converted to the corresponding annual rate.

For ex. a bond with a yield of 4% semi annually, will result in a BEY of (4%*2=)8% annually.

Bond Equivalent Yield And Annual-pay Yield

- The following formula identifies the relationship between BEY and YTM.
- **Bond Equivalent Yield(BEY) of an Annual-pay Bond**

$$BEY = 2 * \left[(1 + \text{Annual YTM})^{1/2} - 1 \right]$$

- **Yield on an annual pay basis**

$$YTM = \left[\left(1 + \frac{\text{BEY}}{2} \right)^2 - 1 \right]$$

Example 3

- What is the yield on a bond equivalent basis of an annual-pay 9% coupon bond priced at par?

a) 4.4%

b) 9%

c) 8.8%

d) 9.5%

Change in Bond Price with change in Yield

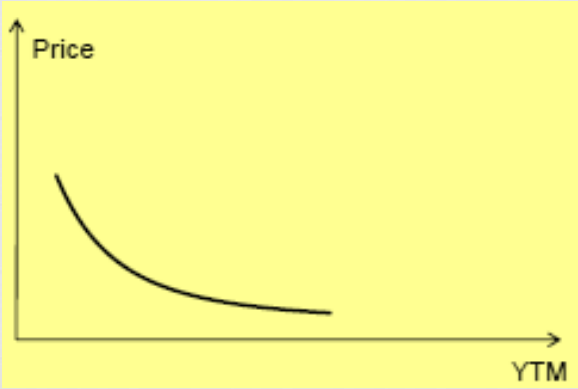
- As the required yield of bonds increases their prices decrease.
- Yield sets the standard for the level of returns to be provided by a bond. If the yield increases, it would mean that a bond that was trading at par prior to this, would now offer less return than required. Thus its price would decrease and similarly for a decrease in yield would cause increase in price.

Change in Bond Price with change in Yield

- This can also be seen from the relation:

$$\text{Bond Price} = \frac{CPN(1)}{1+YTM} + \frac{CPN(2)}{1+YTM^2} + \frac{CPN(3)}{1+YTM^3} + \cdots . + \frac{CPN(n-1)}{1+YTM^{n-1}} + \frac{CPN(n)}{1+YTM^n} + \frac{Par\ Value}{1+YTM^n}$$

- The same can be shown through a price-yield curve:



Duration

- There are 3 possible interpretations of duration:

1.It is the slope of the price-yield curve at the bond's current YTM

2.It is a weighted average of the time until the cash flow will be received. The weights are the proportion of the bond value that each cash flow represents.

3.It is also the approximate change in bond price for a 1% change in yield.

Duration

- Using the third interpretation, the change in price of a bond caused by a change in yield can be approximated as:
- $\Delta P/P = -D^*\Delta Y$

Where,

ΔP is the change in bond price

P is the original bond price

D is the duration of the bond

ΔY is the change in yield

Effective Duration

- Duration is the measure of how long on an average the holder of the bond has to wait before he receives his payments on the bond. A coupon paying bond's duration would be lower than "n" as the holder gets some of his payments in the form of coupons before "n" years
- •In simple words, duration of a bond is sensitivity of bond price to change in its interest rate
- •Effective duration is calculated as:

$$\text{Effective Duration} = \frac{(\text{Bond price when yield falls} - \text{Bond price when yield rises})}{2 * (\text{Initial Price}) * (\text{Change in yield in decimals})}$$

Percentage Change in Price Using Duration

- Approximate percentage price change = - Duration * Δy * 100

- Example: you hold a bond that has a duration of 7.8 years. The interest rates fell by 25 bps. Calculate the approximate percentage price change.

  ➔ Approximate percentage price change =

  - Duration * Δy * 100 = -7.8 *(- .0025) * 100 = 1.95%

- For large changes in yield, convexity should also be used. Percentage change in price becomes inaccurate with only taking duration into account.

Alternative Definitions of Duration

- Macaulay Duration:

  is the weighted average of the times when the payments are made. And the weights are a ratio of the coupon paid at time “t” to the present bond price

Alternative Definitions of Duration

- Macaulay duration is also used to measure how sensitive a bond or a bond portfolio's price is to changes in interest rates.

$$\text{Macaulay Duration} = \frac{\sum_{t=1}^{n} \frac{t * C}{(1+y)^t} + \frac{n * M}{(1+y)^n}}{\text{Current Bond Price}}$$

where:

t = Respective time period

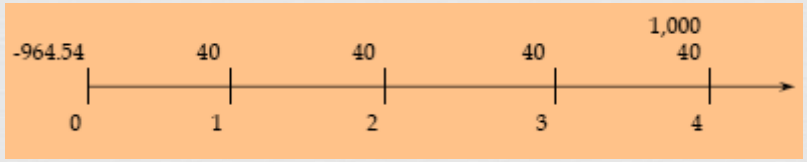
C= Periodic Coupon payments ; y =Periodic yield ;

n = Total number of periods

M = maturity Value

Alternative Definitions of Duration

- Calculating Macaulay Duration:



$$D = \frac{\frac{40}{(1.05)}(1) + \frac{40}{(1.05)^2}(2) + \frac{40}{(1.05)^3}(3) + \frac{1040}{(1.05)^4}(4)}{964.54} = \frac{3636.76}{964.54} = 3.77$$

Note that this is 3.77 six-month periods, which is about 1.89 years

Change In Bond Price With Change In Discount Rate

- Modified Duration

$$\text{ModD} \approx -\frac{1}{\text{P}}\frac{\Delta P}{\Delta y} \Rightarrow \Delta P \approx -P.ModD.\Delta y$$

-The modified duration is equal to the percentage change in price for a given change in yield.

- Example: The current price of a bond is 98.75. Its modified duration is 5.2 years. The YTM of the bond is 7.5%. What would the price be if the yield became 8%?

  $\Delta P = -98.75 * 5.2 * 0.005 = -2.57$

  The new price of the bond is 96.18

Alternative Definitions of Duration

- Modified Duration: is derived from Macaulay Duration. It is better than Macaulay Duration as it takes into account the current YTM.

$$\text{Modified Duration} = \frac{\text{Macaulay Duration}}{\left(1 + \frac{\text{YTM}}{\text{no of interest payments per year}}\right)}$$

Alternative Definitions of Duration

- In summary duration is,

–The first derivative of the price-yield function

–The slope of the price-yield curve.

–A weighted average of the time till the cash flows will be received.(Macaulay Duration)

–The approximate percentage change in price for a 1% change in yield.(Effective Duration)

Duration of a Portfolio

- Duration of a portfolio is the weighted average of the duration of the individual securities in the portfolio.

$$\text{Portfolio Duration} = W_1D_1 + W_2D_2 + \ldots\ldots + W_ND_N$$

- The problem with the above equation is that it holds good only for a parallel shift in the yield curve. This is because securities with different maturities may have different changes in yield.

Price Volatility and Convexity

- We have already seen that the price-yield curve is a negatively sloped and is a curve. This is referred to as convex.



- Properties concerning the price volatility of an option free bond:

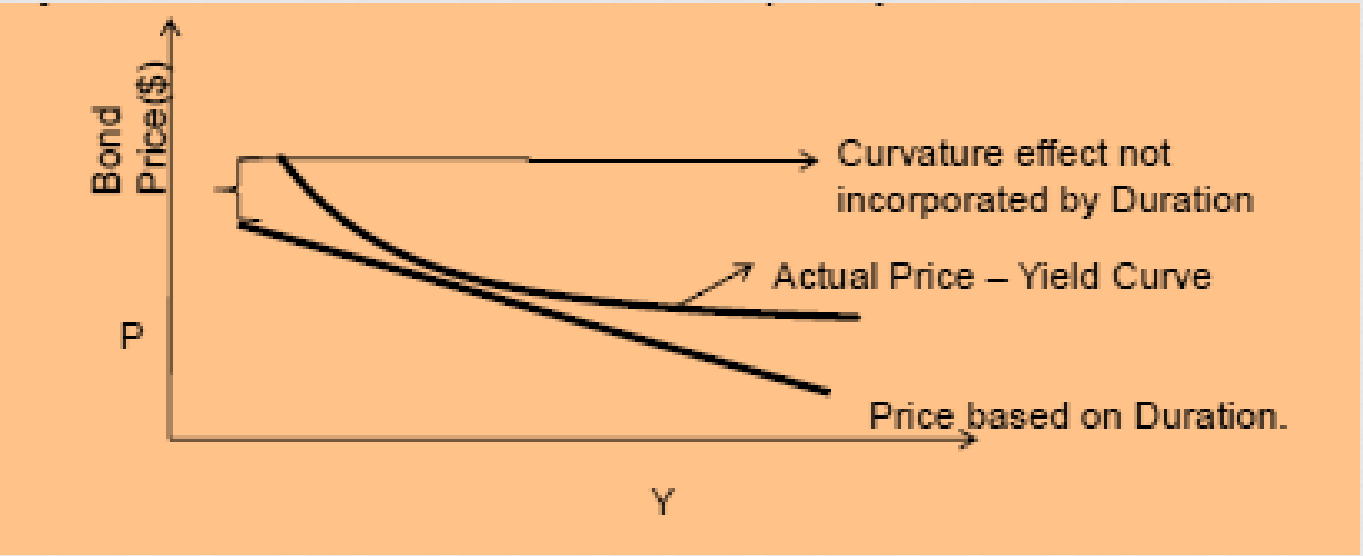
•Percentage price change per change in interest rates is not the same for all bonds

•For either small increases or decreases in yield, percentage change in price for given bond is roughly the same.

•For a given large change in yield, the percentage price increase is greater than the percentage price decrease.

Convexity Measure of a Bond

- Convexity is the measure of the curvature of a price-yield curve.



Convexity Measure of a Bond

- •Duration is an appropriate measure for small changes in the yield. For larger changes in yield convexity should also be used.

  Percentage Change in Price

  = Duration Effect + Convexity Effect

  $=[(-\text{Duration} * \Delta y) + (\text{Convexity} * \Delta y^2)] * 100$

- Note: In this formula all the values are used as numbers.

  E.g. 1% must be written as 0.01.

  This is also the reason to multiply it by 100

Example 4

- A bond has a duration of 7 and a convexity of 65.4. The estimated percentage change in price for this bond, due to a decline in yield of 150 basis points would be?

a) 10.5%

b) -9%

c) 11.97%

d) -10.5%

Price Value of a Basis Point(PVBP)

- This is a measure of interest rate risk.
- PVBP – It is the absolute value of the change in the price of a bond for a 1 basis point change in yield.

PVBP =

|Initial Price - Price when yield changes by 1 basis point|

- The PVBP is the same for both increase and decrease (because change in yield is small)

  PVBP = Duration * 0.01% * Bond Value

Review

$$\text{Value of a bond} = \frac{C}{(1+YTM)} + \frac{C}{(1+YTM)^2} + \frac{C}{(1+YTM)^3} + \ldots\ldots \frac{C+PAR}{(1+YTM)^N}$$

Bond Selling at:	Relationship
Par	Coupon rate = Current Yield = Yield to Maturity
Discount	Coupon rate < Current Yield < Yield to Maturity
Premium	Coupon rate > Current Yield > Yield to Maturity

$$\text{Absolute Yield Spread} = \text{Yield on Bond} - \text{Yield on Benchmark Bond}$$

$$\text{Relative yield spread} = \frac{\text{Absolute yield spread}}{\text{Yield on the benchmark bond}}$$

$$\text{Yield Ratio} = \frac{\text{Subject bond yield}}{\text{benchmark bond yield}}$$

$$BEY = 2 * \left[(1 + \text{Annual YTM})^{1/2} - 1\right]$$

$$YTM = \left[\left(1 + \frac{BEY}{2}\right)^2 - 1\right]$$

Review

$$\text{Modified Duration} = \frac{\text{Macaulay Duration}}{\left(1 + \frac{\text{YTM}}{\text{no of interest payments per year}}\right)}$$

$$\text{ModD} \approx -\frac{1}{V}\frac{\Delta V}{\Delta y} \Rightarrow \Delta V \approx -V\,ModD\,\Delta y$$

$$\text{PVBP} = \text{Duration} * 0.01\% * \text{Bond Value}$$

$$\text{Effective Duration} = \frac{(\text{Bond price when yield falls} - \text{Bond price when yield rises})}{2 * (\text{Initial Price}) * (\text{Change in yield in decimals})}$$

$$\text{Convexity} = \frac{(\text{Bond price when yield falls} + \text{Bond price when yield rises} - 2 * \text{Initial Bond Price})}{2 * (\text{Initial Price}) * (\text{Change in yield in decimals})^2}$$